

across a vast array of skills. By using GPT-4, we were able to efficiently label a wide range of skills, allowing for a comprehensive and consistent categorization of all job postings in the online labor market. We minimized errors using state-of-the-art prompt engineering, measuring accuracy with a test set, and performing manual checks.

In addition, this paper focuses on the partial equilibrium effects of new technology and does not capture potential productivity increases outside of the platform under study. Technological improvements can decrease worker demand, but productivity gains can lead to new capital and organizational investments, potentially increasing labor demand (Acemoglu and Restrepo (2018), Autor and Salomons (2018)). To better understand how substitution of freelancers with GenAI tools affects in-house labor, one would need to model the fragmentation of production into tasks in more detail (Rubbo, 2023).

A further limitation is that we only observe labor demand but not wages. Thus, we cannot definitively attribute changes in job postings to shifts in the demand curve. Workers might decrease their wage bids if they face more competition, or they might increase their wage bids if they believe ChatGPT enhances their value. However, previous research (Dube et al., 2020; Duch-Brown et al., 2022; Horton, 2021) suggests that online labor market workers have low bargaining power and act as price-takers. Moreover, our analysis indicates that our results are not substantially biased by simultaneity issues arising from labor supply responses to the ChatGPT launch. Specifically, we demonstrate that while the launch of ChatGPT led to a decrease in labor demand, average employer budgets remained unchanged. Additionally, we observe an increase in the number of applicants per job posting is of the same magnitude than the reduction in job postings. These findings collectively support the conclusion that ChatGPT's launch led to reduced labor demand, with workers competing for a smaller pool of substitutable jobs.

While this analysis focuses on the short-term impacts of LLMs, understanding the long-term effects requires examining how these impacts vary based on workers' skills (Teutloff et al., 2023; Mealy et al., 2018), location (Farinha et al., 2020; Braesemann et al., 2022), and the alignment between current skills and those needed for emerging jobs (Neffke et al., 2024; Neffke and Henning, 2013). Our results provide evidence of first-order demand changes, which can be integrated into more complex models of worker adaptation (del Rio-Chanona et al., 2021; Acemoglu and Autor, 2011) to answer the broader question on the long-term effects of LLMs. Future research could examine the timing of demand shifts more precisely, investigating how long it takes employers to incorporate AI-driven tools and what the transition period looks like. Looking more at the firm side could reveal more about how to generalize from findings on contract work platforms to traditional labor markets. In addition, further work could explore whether the tasks whose demand decreased are now performed more effectively, performed differently, or not performed at all.

Our study reveals that a greater proportion of skill clusters are complemented by AI rather than substituted. However, while some of these complementary skill clusters experienced an increase in demand, others saw a decline. This underscores the critical role of public and private decision-makers in shaping the socio-economic impact of AI (Ritala et al., 2023; Gerling et al., 2024). Managers and policy-makers need to support workers in adapting and reskilling and should empower them to use LLMs in complementary ways (Retkowsky et al., 2024; Mollick et al., 2024). Our findings from online labor markets, which adapt quickly to technological changes, provide insights into potential future effects on traditional labor markets. Although traditional markets may adapt more slowly due to contract and severance laws, they will likely follow similar patterns over time. Policymakers must implement reskilling and upskilling programs to help workers transition into roles that complement AI technologies, focusing on foundational skills (Hosseinioun et al., 2023) and analytical thinking (Humburg and Van der Velden, 2017). Overall, our study demonstrates that while AI tools like ChatGPT can substitute some jobs, it also drives the development of new products and services, exemplifying creative destruction (Klimek et al., 2012) and contributing to the economy's evolution towards greater complexity (Balland et al., 2022).

Statements

Declaration of generative AI in scientific writing

During the preparation of this work the authors used ChatGPT 4.0 in order to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Online appendix: supplementary analyses

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jebo.2024.106845>.

Data availability

The authors do not have permission to share data.

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